



NASDAQ Index Price Prediction

207 Section 3 - Spring 2025 - Final Project

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Motivation





Main Question

- Do macroeconomic indicators improve NASDAQ predictions in an LSTM model?



What Has Been Done To Date?



■ Good Results On Long Term Predictions

[Home](#) > [Proceedings of the 2nd International Conference on Business and Policy Studies](#) > Conference paper

Inflation-Related Factors Enhanced LSTM-Based Multivariate Time Series for Stock Market Forecasting

Conference paper | First Online: 08 October 2023

pp 72–82 | [Cite this conference paper](#)

This paper found CPI, PPI, M2 are key to improving SSE index predictions

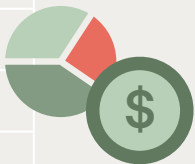
Macroeconomic Indicators Alone can Predict the Monthly Closing Price of Major U.S. Indices: Insights from Artificial Intelligence, Time-Series Analysis and Hybrid Models

July 2018 · [Applied Soft Computing](#) 71(1)

DOI:[10.1016/j.asoc.2018.07.024](#)

And this paper found CPI, PPI, M2 have a statistically significant impact on long term predictions





Motivation

Can macroeconomic indicators improve NASDAQ predictions?

- Stock market models are everywhere—but curiously, few use macroeconomic data. We wondered why.

What's been done:

- Most models rely on past prices and technical indicators. **Macroeconomic features** like interest rates and inflation are rarely used, likely due to reporting delays and added complexity.

Our approach:

- We combined 5 years of stock and macroeconomic data, engineered features, and trained LSTM models to compare performance with and without macro inputs.

What we found:

- Models using only stock price history **outperformed** those using macroeconomic data.



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Data



Dataset



Source

- Yahoo Finance (yfinance)
- Federal Reserve Economic Data (FRED)

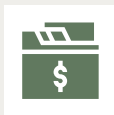


Build

- Fetch NASDAQ data from yfinance
- Fetch macro data from FRED
- From 1/1/2020 - 12/31/2024



Data Pre-Processing



Macro Shifts

- FEDFUNDS - 30 days
- DGS10 - 1 day
- CPI/PPI - 45 days
- M2SL - 60 days
- After shifting, we forward-filled missing values



Data Splits

- Train: 2020-2022
- Validate: 2023
- Test: 2024



Sequences

For each prediction in our dataset, we:

- Took a window of n_steps past days as our input feature
- Use the next day's target as our label

Example:

1 2 3 4 5 6 7 8 9 10 11 12 13 14 30

Label
31

Input Features

Example: If $n_steps = 30$, the model learns to predict what happens on day 31 based on data from days 1-30. Then it shifts the window by one day and repeats.



Features - Yahoo Finance



Variable Name(s)	Description
Close	Predicted variable; daily price of NASDAQ index at market close
Open	Daily price of NASDAQ index at market open
High, Low	Highest, lowest daily price of NASDAQ index (respectively)
Volume	Number of trades per day for NASDAQ index
TLT_Open, TLT_Close	Daily open and close prices of the iShares 20+ Year Treasury Bond ETF (TLT), which tracks the ICE U.S. Treasury 20+ Year Bond Index; reflects investor sentiment on long-term U.S. Treasury yields
TLT_High, TLT_Low	Highest and lowest daily prices of the iShares 20+ Year Treasury Bond ETF (TLT)
TLT_Volume	Number of trades per day for the iShares 20+ Year Treasury Bond ETF (TLT)
VIX_Close	Closing price of the CBOE Volatility Index; often referred to as the “fear index” because it spikes when market uncertainty or risk increases



Features - FRED + Engineered



Source	Variable Name	Description
Federal Reserve Economic Data	FEDFUNDS	Federal Funds Rate; the short-term interest rate set by the Fed
(FRED)	DGS10	10-Year Treasury Yield; long term market rate for gov bonds
(FRED)	CPIAUCSL	Consumer Price Index; measures inflation
(FRED)	PPIACO	Producer Price Index; tracks price changes at the producer level
(FRED)	M2SL	M2 Money Supply; tracks liquidity (available money in circulation)
Engineered	MACD	Moving Average Convergence Divergence; compares short & long-term moving averages of price
Engineered	MACD_Signal	Signal line for MACD; a 9-day EMA of MACD used to generate buy/sell signals
Engineered	MACD_Hist	MACD histogram; the difference between the MACD line and the signal line
Engineered	Yield_Spread	Difference between DGS10 and FEDFUNDS; used to gauge the slope of the yield curve



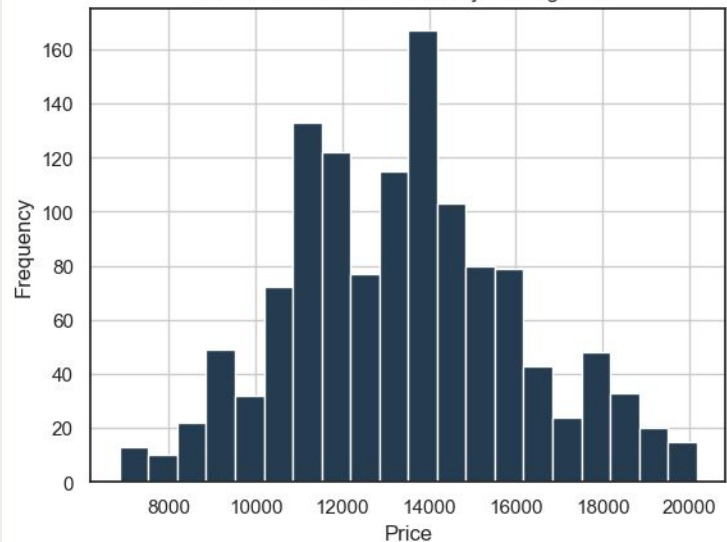
Exploratory Data Analysis



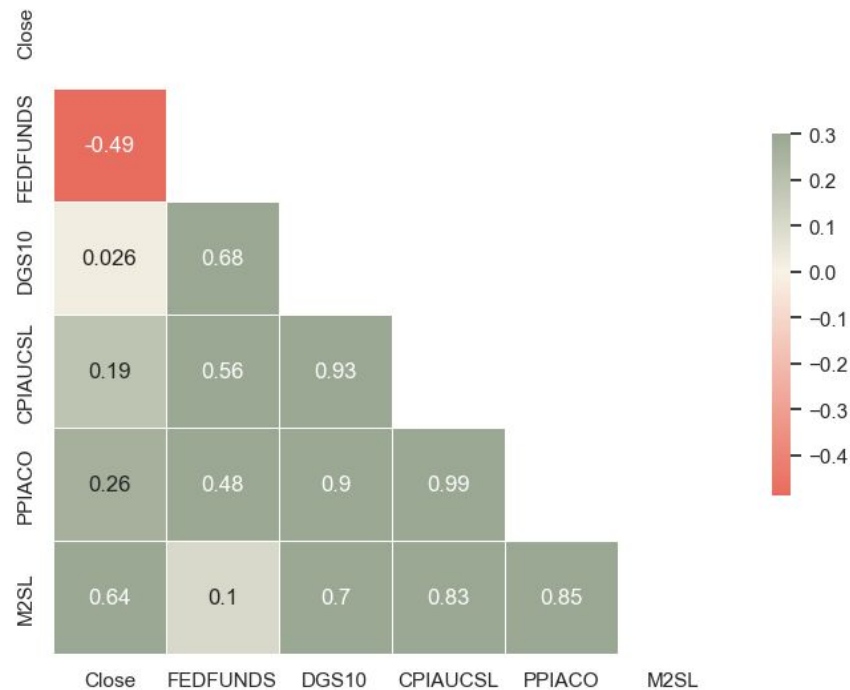
NASDAQ Index - Closing Prices (2020-2024)



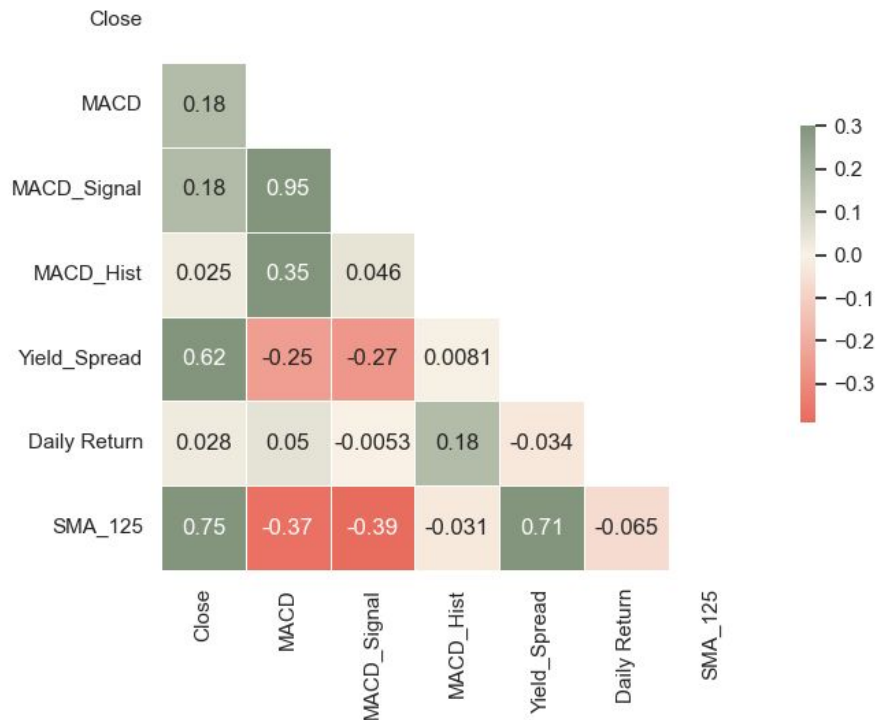
Distribution of NASDAQ Daily Closing Price



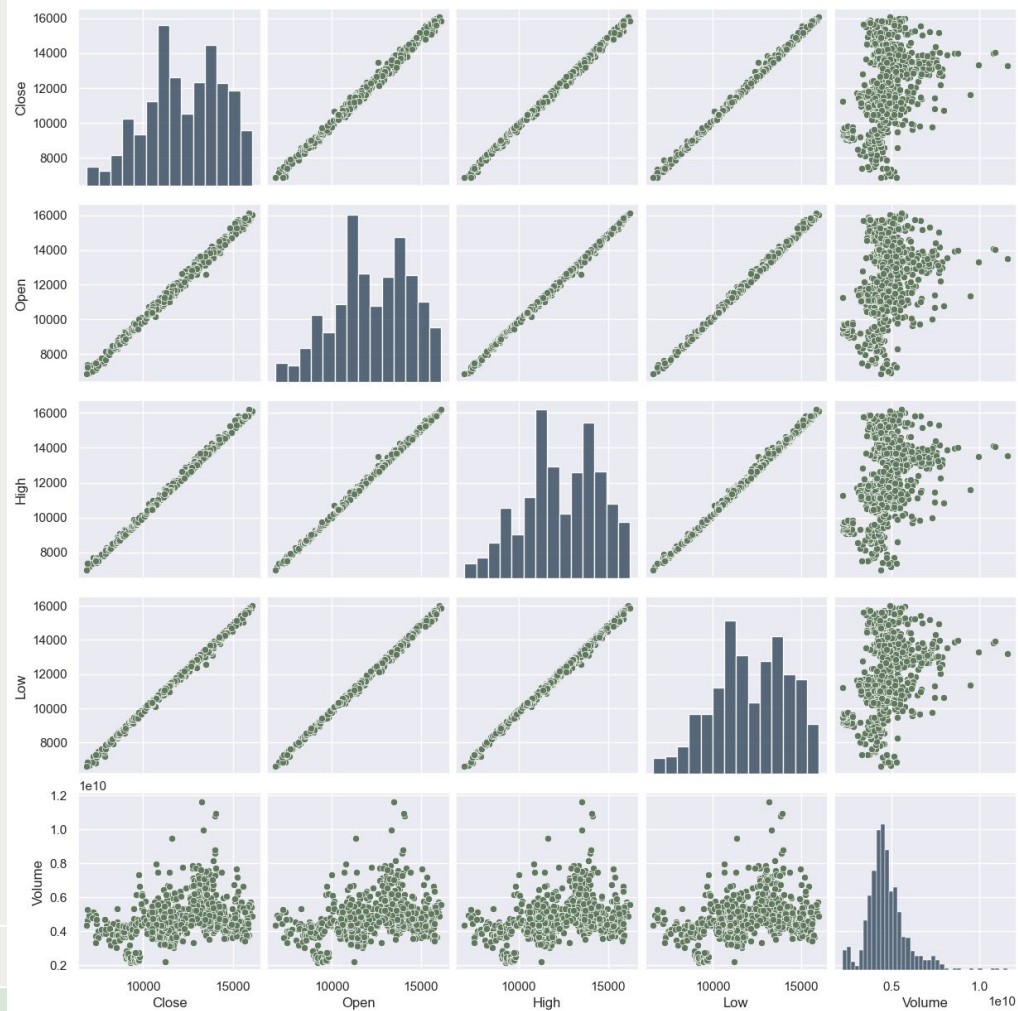
Correlation Between Closing Price and Macroeconomic Indicators



Correlation Between Closing Price and Engineered Features



Scatterplot Matrix of NASDAQ Features (Training Data)

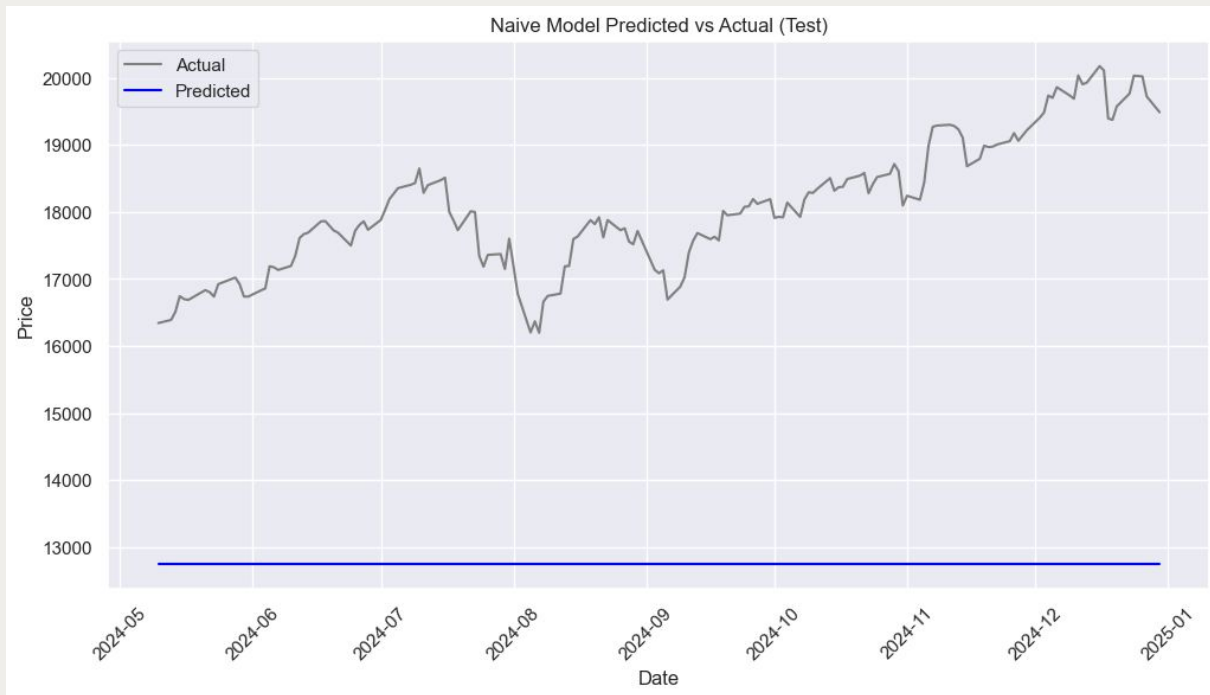


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Modeling



Baseline Naïve Model



Model Structure:

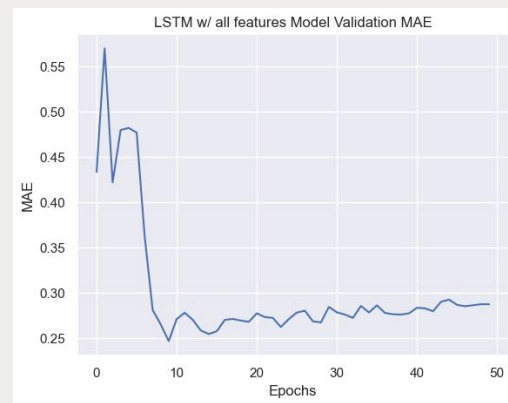
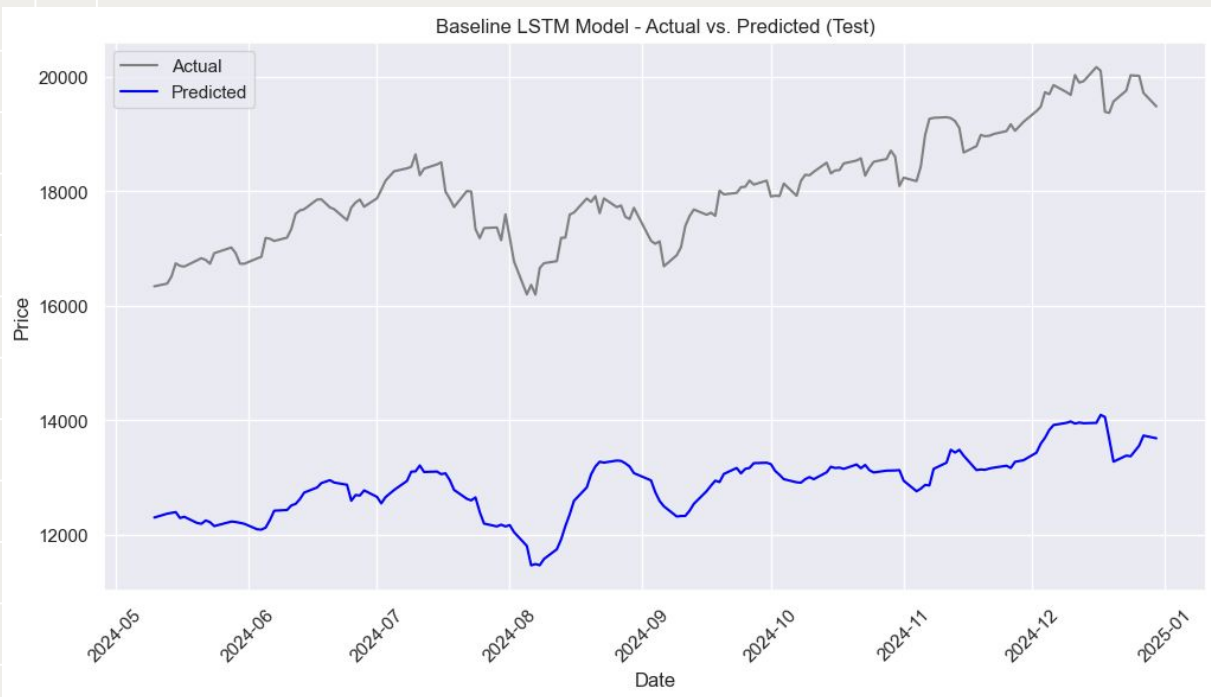
Always predict the average price of the training set

Val MAE: \$978.37

Test MAE: \$5,298.17



Baseline LSTM Model - All Features



Features: All yfinance, FRED, and engineered features

Layer: 1

Units: 32

Learning Rate: 0.001

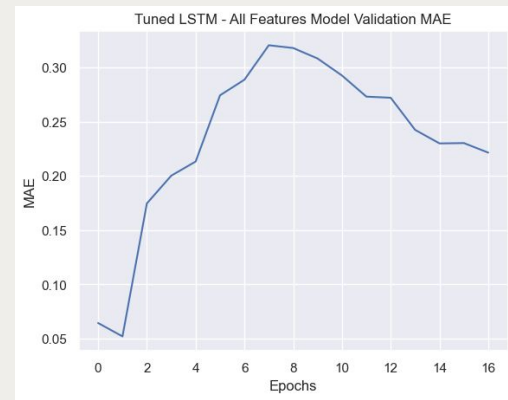
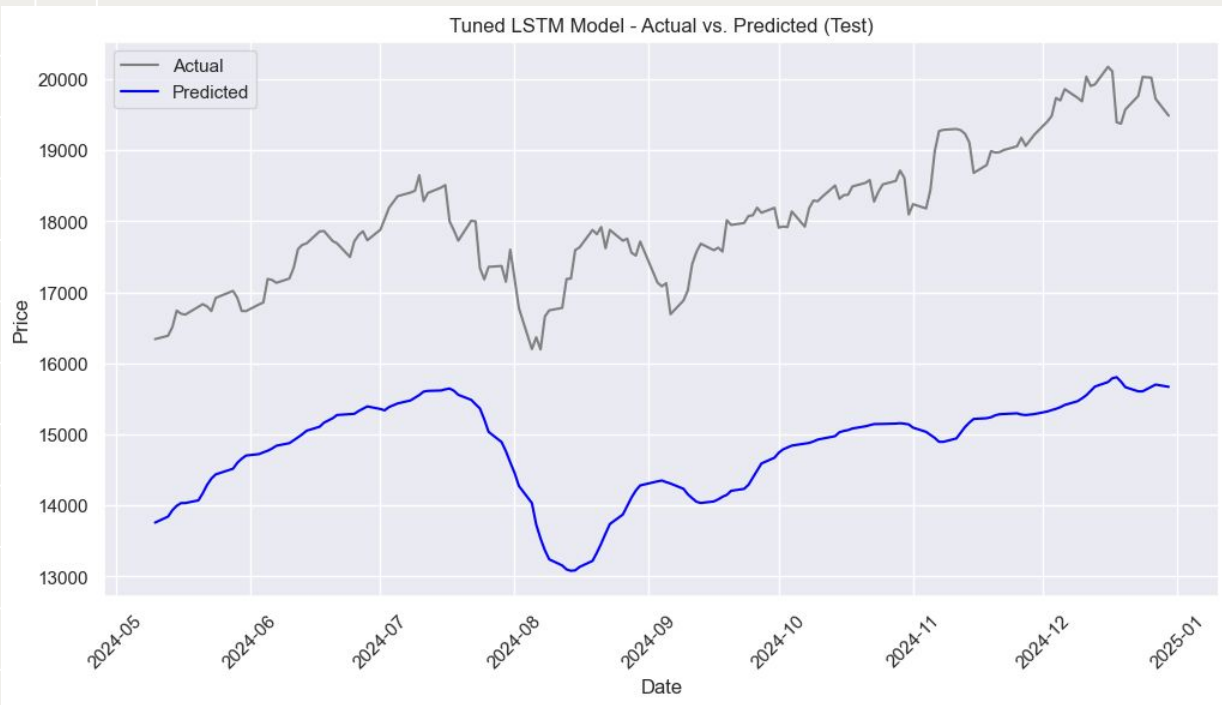
Epochs: 50

Val MAE: \$2,643.66 (28.75% within range)

Test MAE: \$5,174.76 (56.27% within range)



Tuned LSTM Model - All Features



Features: All yfinance, FRED, and engineered features

Layer: 1

Units: 32

Learning Rate: 0.005

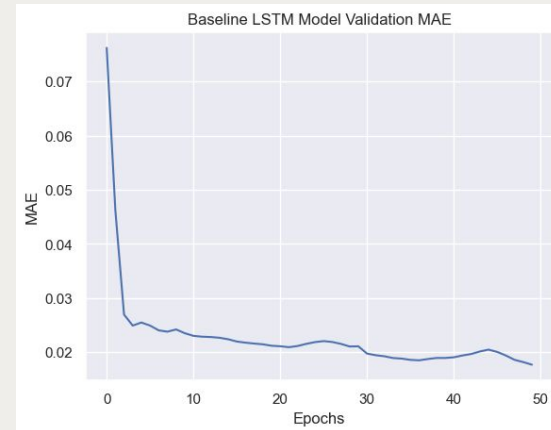
Epochs: 16

Val MAE: \$477.50 (5.19% within range)

Test MAE: \$3,251.96 (35.36% within range)



Baseline LSTM Model



Features: Previous 90 Days' Open, Low, High, Volume

Layer: 1

Units: 32

Optimizer: Adam

Learning Rate: 0.001

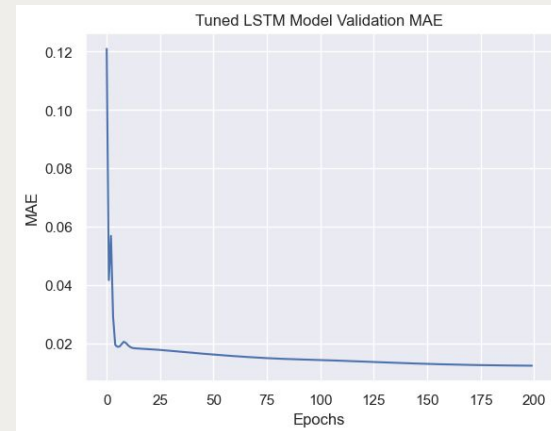
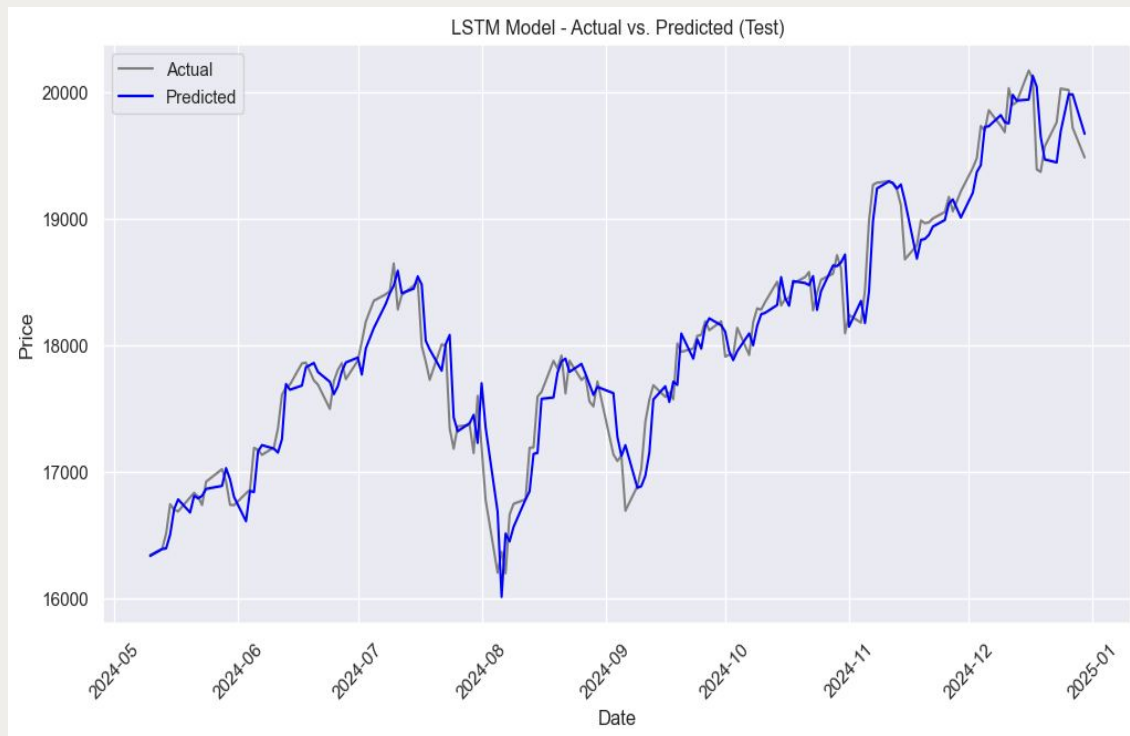
Epochs: 50

Val MAE: \$162.47 (1.77% within range)

Test MAE: \$228.62 (2.49% within range)



Tuned LSTM Model

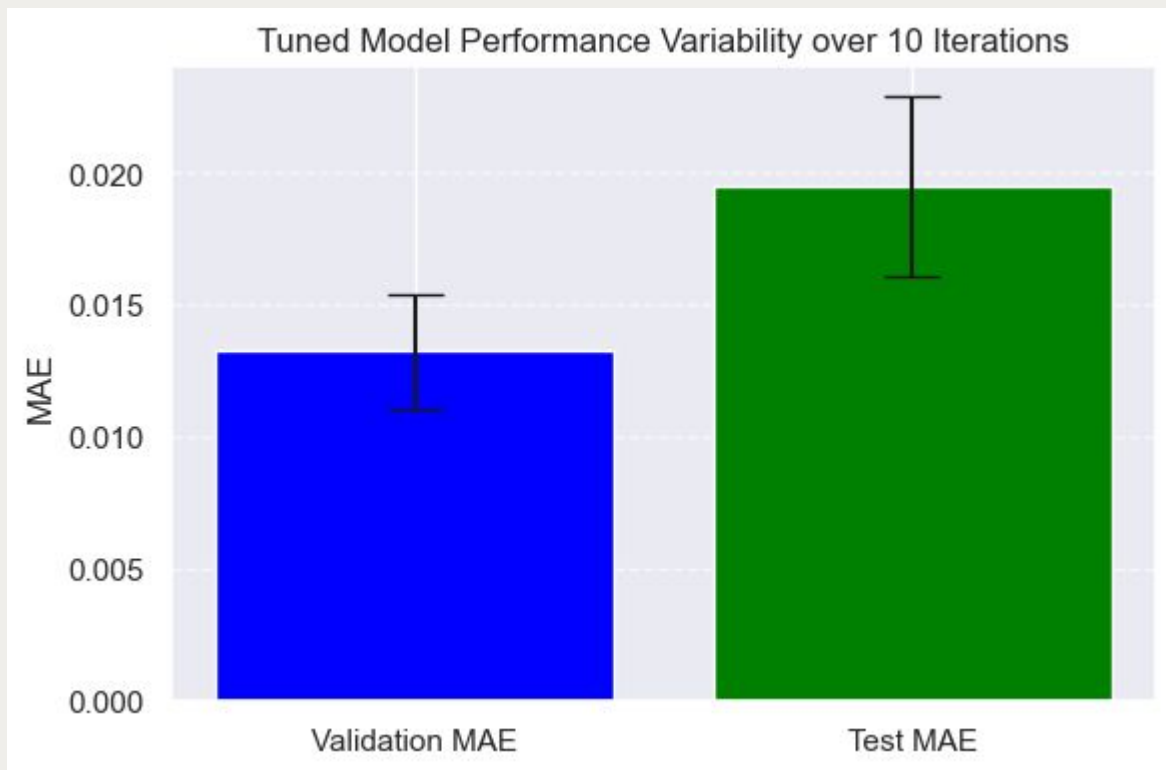


Features: Previous 90 Days' Low, High
Layer: 1
Units: 64
Optimizer: Adam
Learning Rate: 0.01
Epochs: 200

Val MAE: \$113.11 (1.23% within range)
Test MAE: \$159.78 (1.74% within range)



Tuned LSTM Model Variability



Val MAE Mean: 0.013
(MinMaxStandardized)

Test MAE Mean: 0.019
(MinMaxStandardized)

Val MAE Mean Std: 0.0023
Test MAE Mean Std: 0.0036



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Experiments





Number of LSTM layers

■ 1 layer

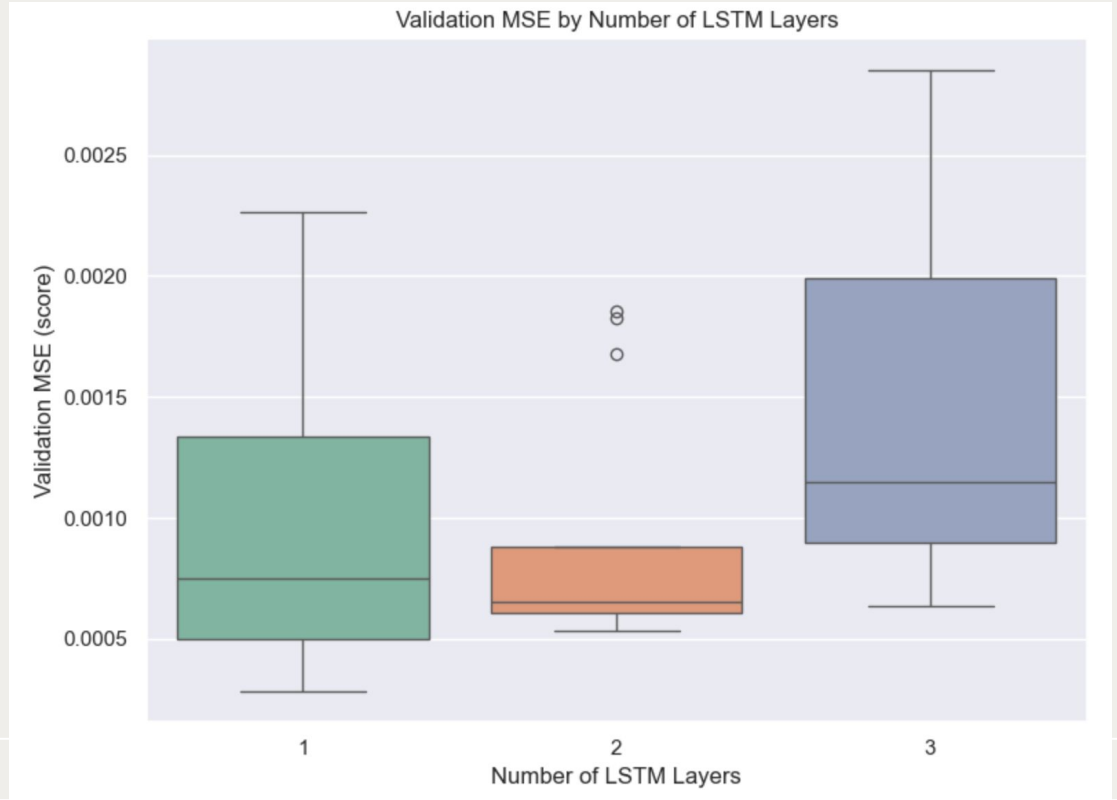
No significant outliers

■ 2 layers

Lowest median value
Noticeable outliers

■ 3 layers

Highest median value



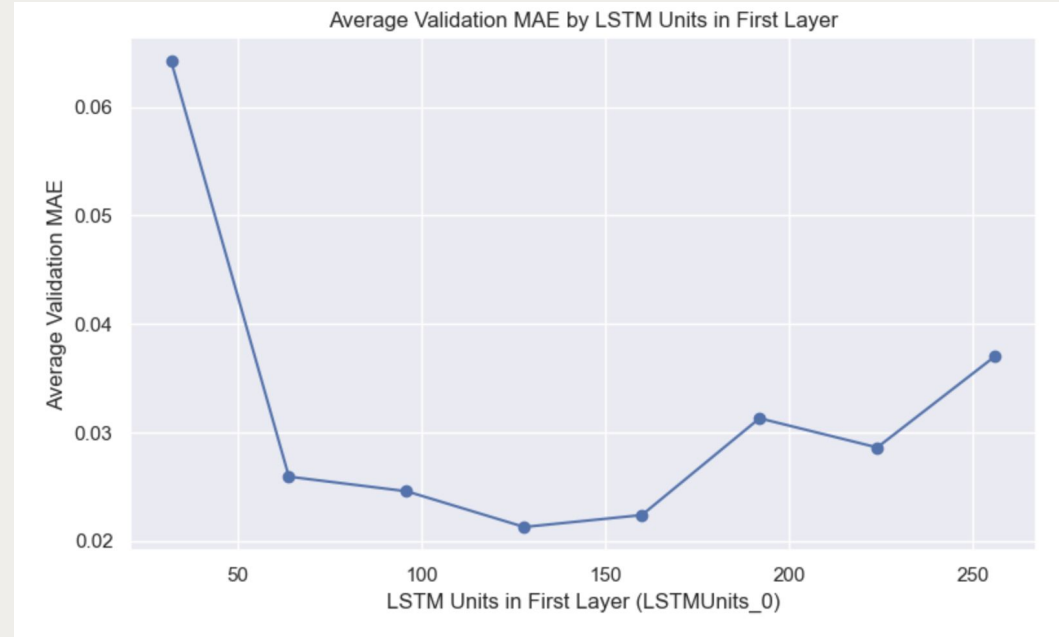


Number of Neurons

■ Diminishing Return

Significant drop of performance as neurons increase up until 125 neurons

Larger number of neurons does not show continuous improvement in model performance



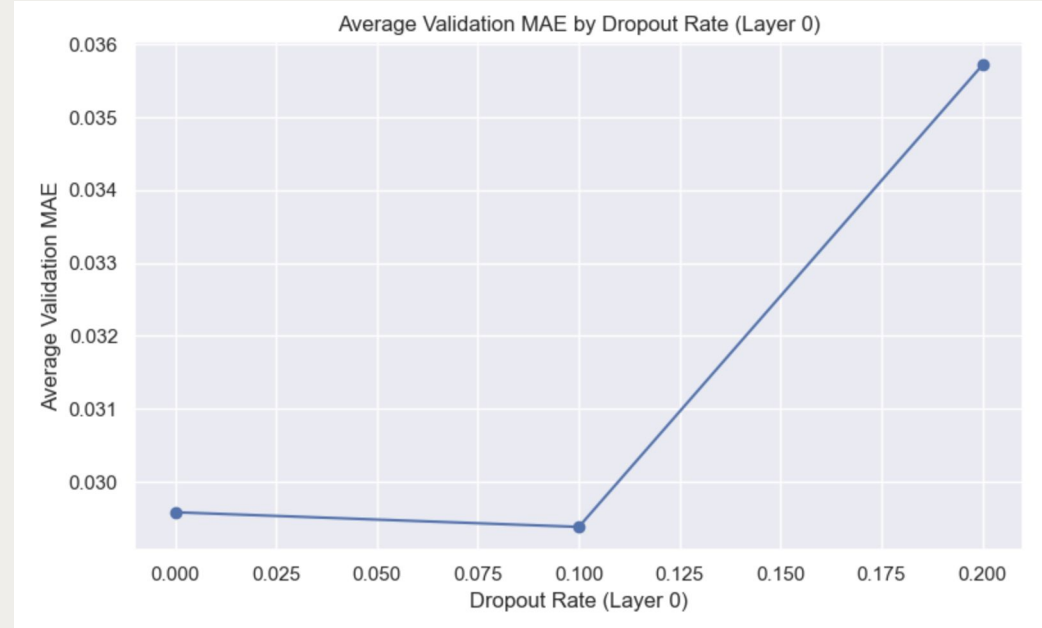


Dropout Rate

■ Less is more

Best performance with 0.1 dropout rate

Sharp increase in MAE for higher rates



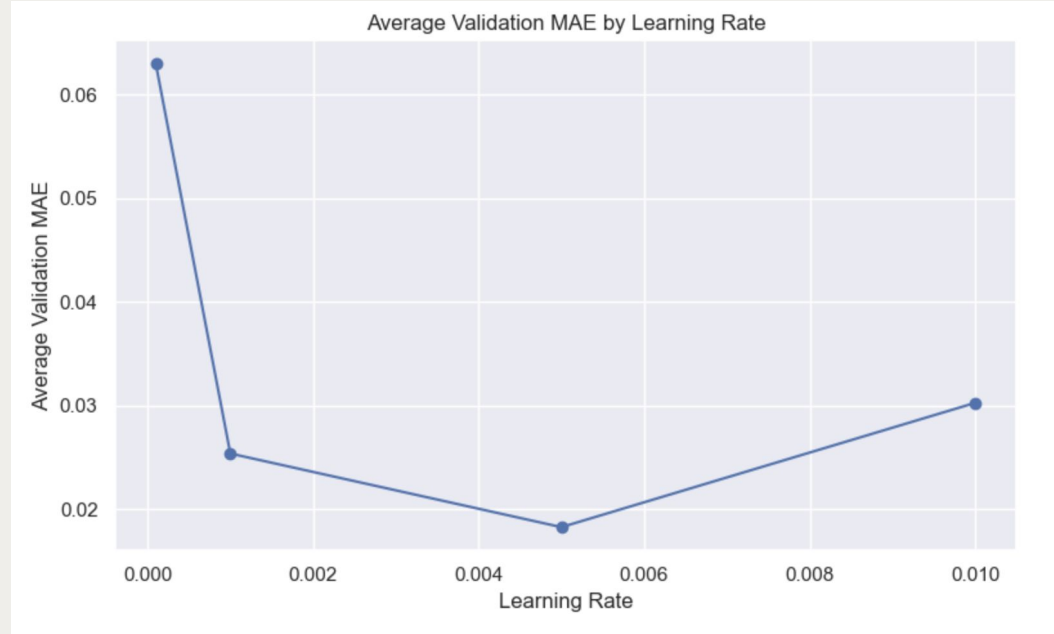


Learning Rate

■ The sweet spot

Improvement in performance ease out as learning rate increases

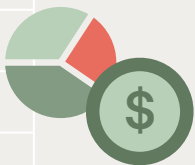
0.005 shown the best performance



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Conclusions





Macro Data vs Long Term Data

Findings

1. Be selective! As number of features and model complexity increased, performance tended to decrease
 - The team ran the model with different features and hyperparameter configurations, including all features, only yfinance features, and only engineered features
 - Our fine-tuned model with two long term features and one LSTM layer performed the best
 - The model run with all features converged at 0.1 MAE, while our fine-tuned model converged around 0.02 MAE
2. Long term metrics improve the output. We found that the 90 day low and high price features provide the most accurate predictions



If we had more time...

Hybrid Models

We would explore combining LSTMs with attention mechanisms or transformers to better capture relevant patterns in mixed data types.

Model Generalization

We would extend the analysis to other indices (e.g., S&P 500, Dow Jones) to test robustness.



Thanks!



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NeurIPS Checklist



Societal Impact

Intended Use:

- Our project is an **academic exploration** of time-series forecasting using deep learning to predict the NASDAQ index.
- It is **not intended** for live deployment or financial decision-making.

Financial Misuse:

- Even with disclaimers, models like ours could be repurposed to support automated trading strategies that **mislead** retail investors, especially if performance claims are overstated or misunderstood.

Market Bias Reinforcement

- Relying solely on historical financial data and macroeconomic indicators could bake in **systemic biases**, disproportionately affecting groups with less market access.



Limitations

Approach Limitations:

- Our model was trained and tested on a single dataset: the NASDAQ index, with closing price as the lone target variable.
- Our results do not apply to other indices or asset classes, which will have their own unique contexts.

Model Assumptions:

- LSTM models assume a consistent temporal pattern in the data—highly volatile environments could reduce accuracy.
- Our model does not account for exogenous shocks (e.g., COVID-19, wars), which can dominate market movement.



NeurIPS Checklist



Checklist Item	Response
Do the main claims made in the abstract and introduction accurately reflect the presentation's contributions and scope?	Yes
Did you discuss any potential negative societal impacts of your work?	Yes
Did you describe the limitations of your work?	Yes
Did you include the code, data, and instructions needed to reproduce the main experimental results?	Yes
Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)?	Yes
Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? Answer "yes" if you report error bars, confidence intervals, or statistical significance tests for your main experiments.	Yes (see notebook)
Did you include the amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)?	All training was done on our local machines
If your work uses existing assets, did you cite the creators and the licenses?	Yes



Team & Contributions

All Team Members

- Exploratory Data Analysis
- Data Visualization
- Slides

Clara Rhoades

- Data processing
- Baseline modeling

Isaac Dallas

- Feature engineering

Coco Sun & Yanling Liu

- Model experimentation
- Hyperparameter tuning



References

- R. Lau, yfinance, Version 0.2.36, 2024. [Online]. Available: <https://github.com/ranaroussi/yfinance>
- Federal Reserve Bank of St. Louis, FRED data accessed via pandas_datareader, Version 0.10.0, 2024. [Online]. Available: <https://fred.stlouisfed.org/>
- Du, W. et al. (2023). "Inflation-Related Factors Enhanced LSTM-Based Multivariate Time Series for Stock Market Forecasting." (CONF-BPS 2023) – found CPI, PPI, M2 are key to improving SSE index predictions (https://link.springer.com/chapter/10.1007/978-981-99-6441-3_7#citeas)
- Barth, J.R., Weng, B., et al. (2019). "Macroeconomic Indicators Alone can Predict the Monthly Closing Price of Major U.S. Indices." Applied Soft Computing, showed macro features improve index forecasts
(https://www.researchgate.net/publication/326378870_Macroeconomic_Indicators_Alone_can_Predict_the_Monthly_Closing_Price_of_Major_US_Indices_Insights_from_Artificial_Intelligence_Time-Series_Analysis_and_Hybrid_Models)
- We occasionally used ChatGPT to assist with syntax and debugging. OpenAI, ChatGPT, Version 4o (March-April 2025). Accessed via <https://chat.openai.com>
- This presentation template was created by Slidesgo (<https://slidesgo.com/>) and includes icons by Flaticon (<https://www.flaticon.com/>) and infographics and images by Freepik (<https://www.freepik.com/>).

